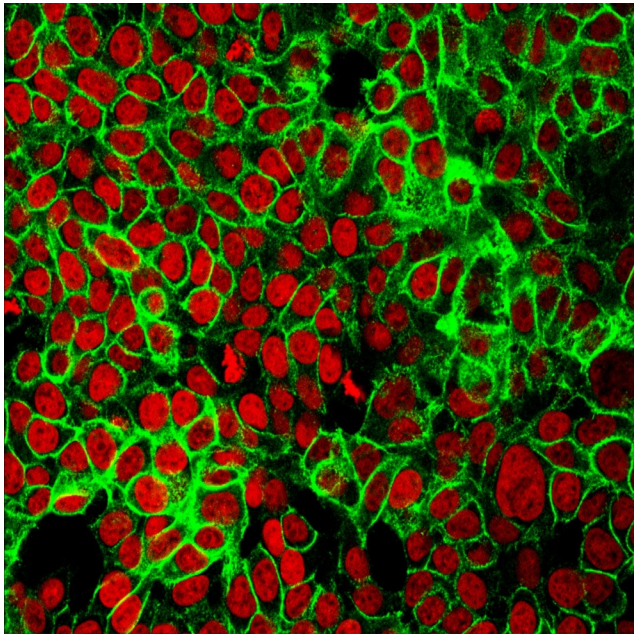
Unsupervised Learning

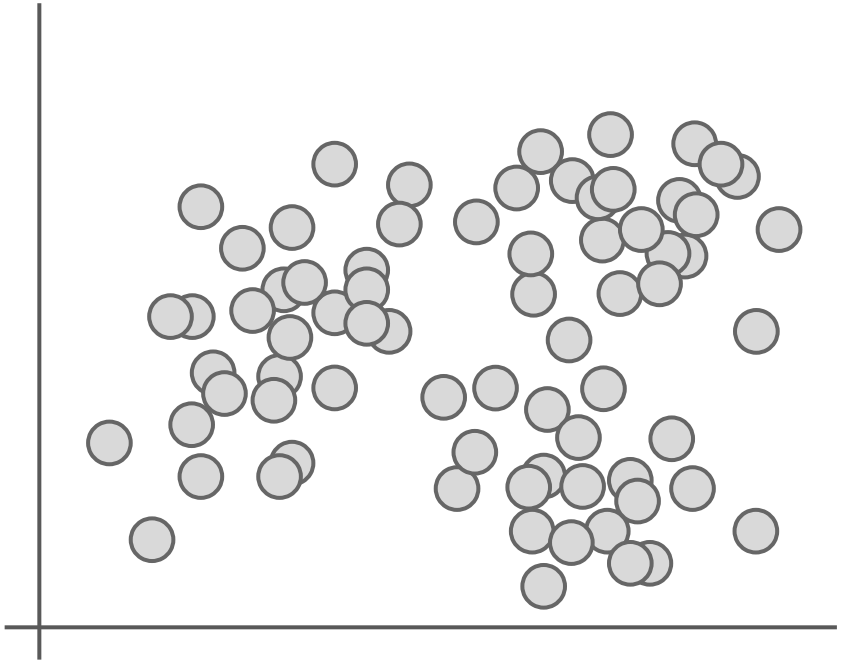
- No labels!
- Model automatically discovers patterns in the data
- Uses
  - Segmentation
  - Clustering
  - Dimensionality reduction
  - Anomaly detection



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K-means Clustering

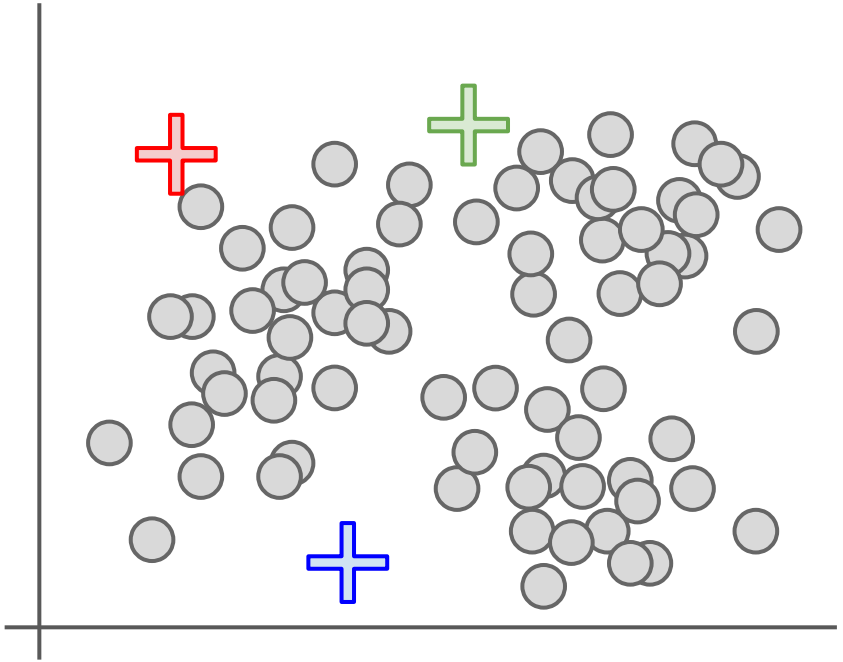
1. Define k (e.g. k=3)



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K-means Clustering

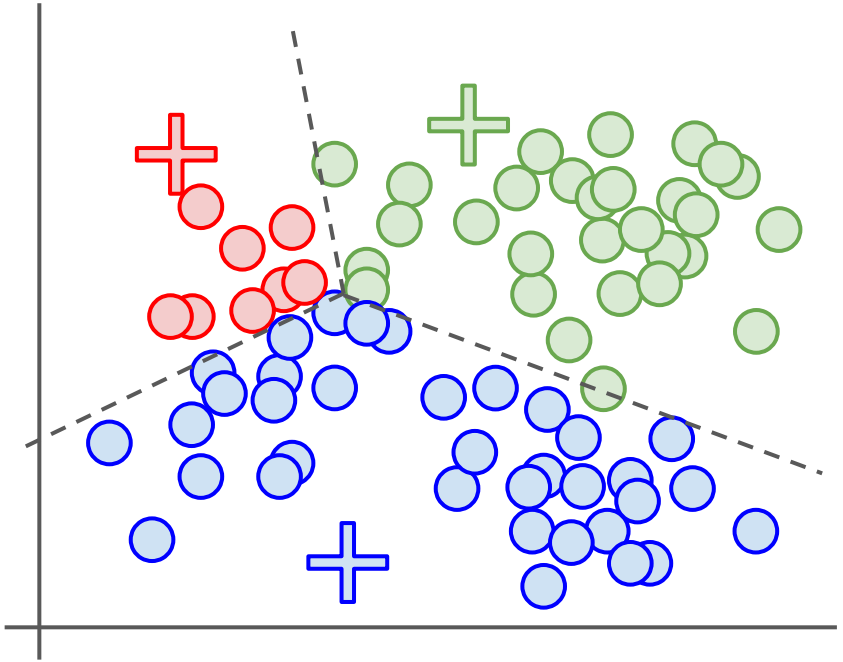
1. Define k (e.g. k=3)
2. Randomly choose centroid for each cluster



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K-means Clustering

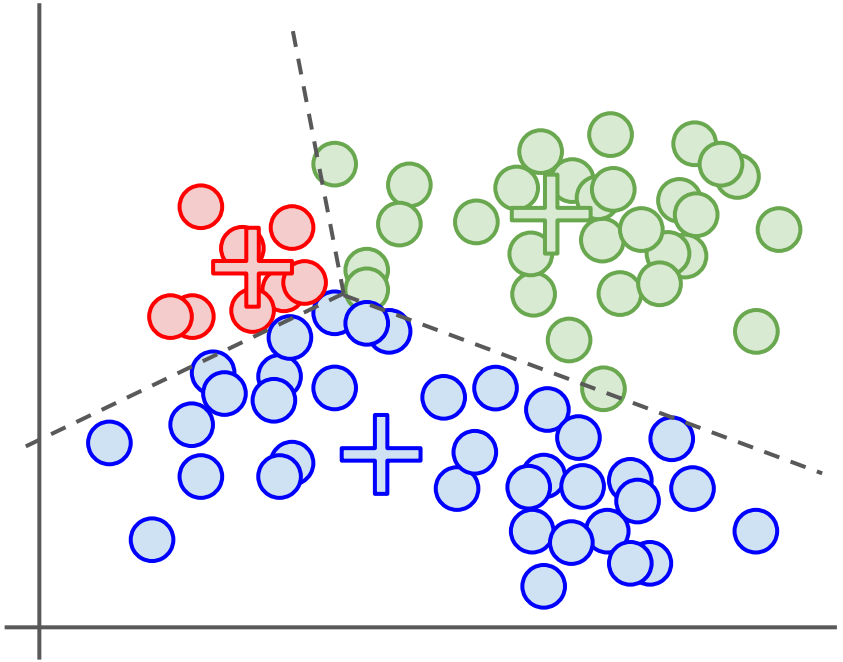
1. Define k (e.g. k=3)
2. Randomly choose centroid for each cluster
3. Assign every sample to nearest centroid based on Euclidean distance



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K-means Clustering

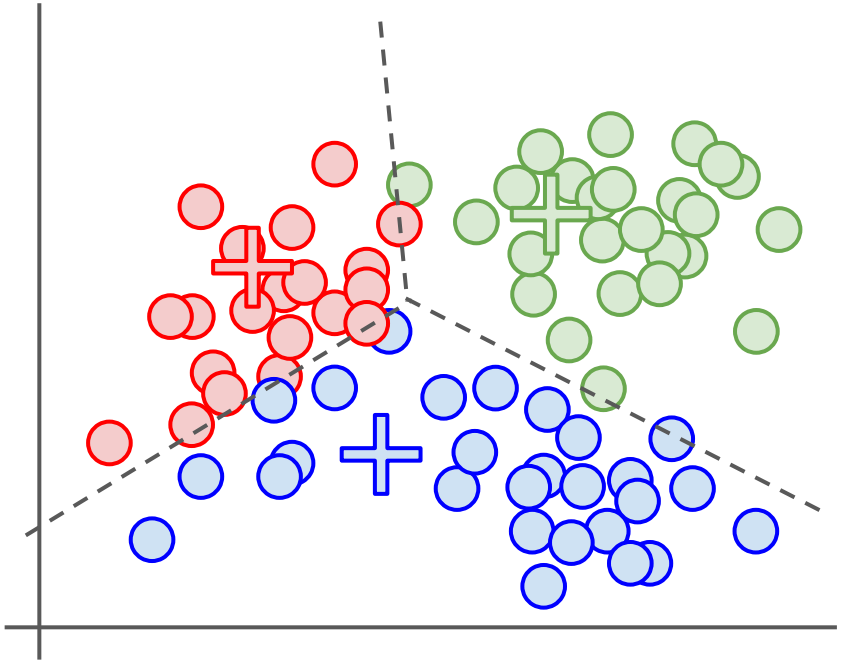
1. Define k (e.g. k=3)
2. Randomly choose centroid for each cluster
3. Assign every sample to nearest centroid based on Euclidean distance
4. Re-compute the centroid of the cluster



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K-means Clustering

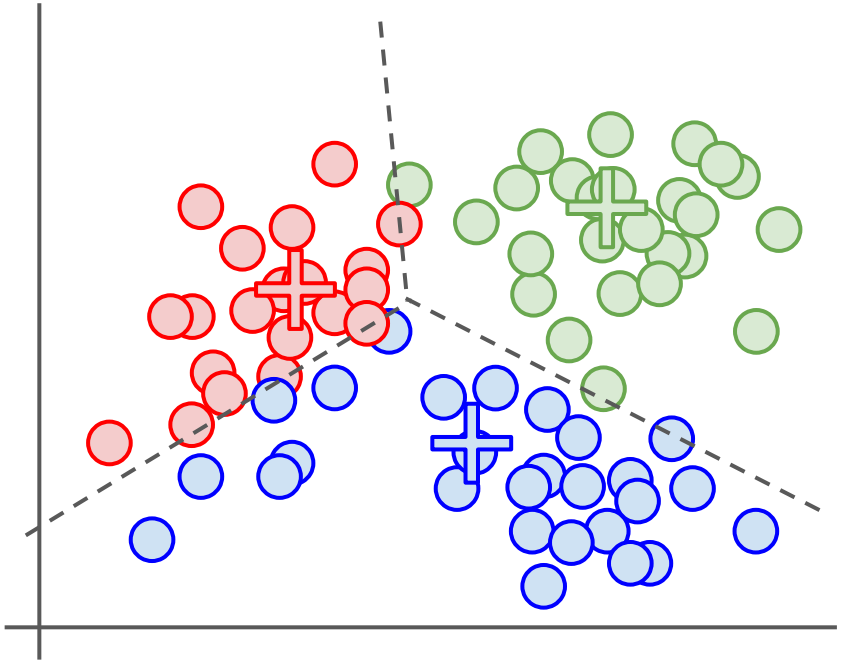
1. Define k (e.g. k=3)
2. Randomly choose centroid for each cluster
3. Assign every sample to nearest centroid based on Euclidean distance
4. Re-compute the centroid of the cluster
5. Repeat steps 3-4



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K-means Clustering

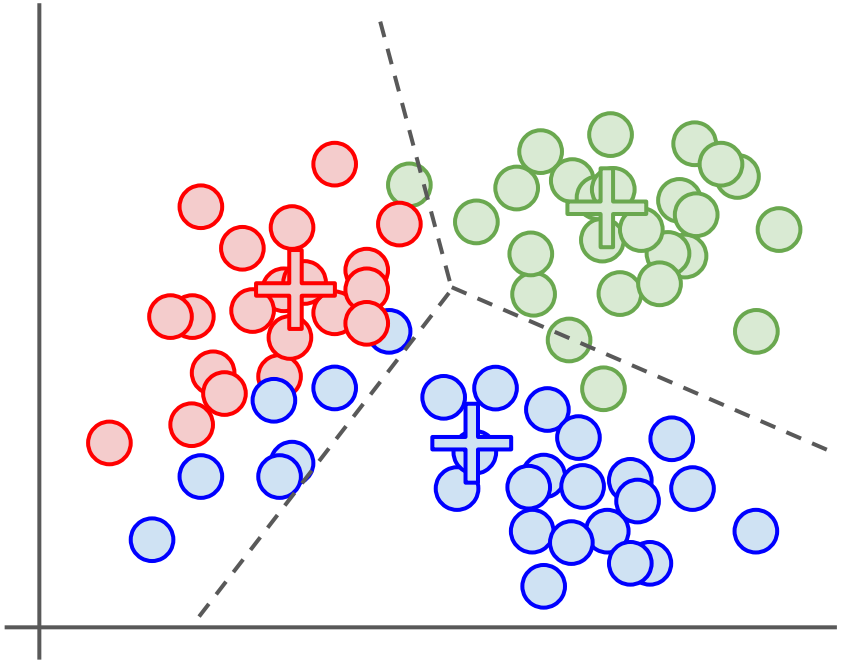
1. Define k (e.g. k=3)
2. Randomly choose centroid for each cluster
3. Assign every sample to nearest centroid based on Euclidean distance
4. Re-compute the centroid of the cluster
5. Repeat steps 3-4



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K-means Clustering

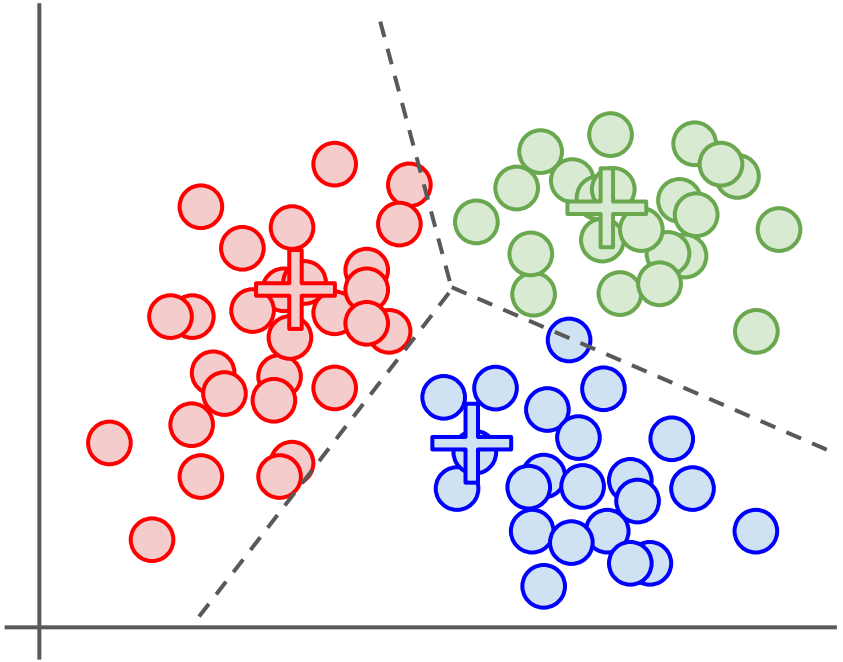
1. Define k (e.g. k=3)
2. Randomly choose centroid for each cluster
3. Assign every sample to nearest centroid based on Euclidean distance
4. Re-compute the centroid of the cluster
5. Repeat steps 3-4



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K-means Clustering

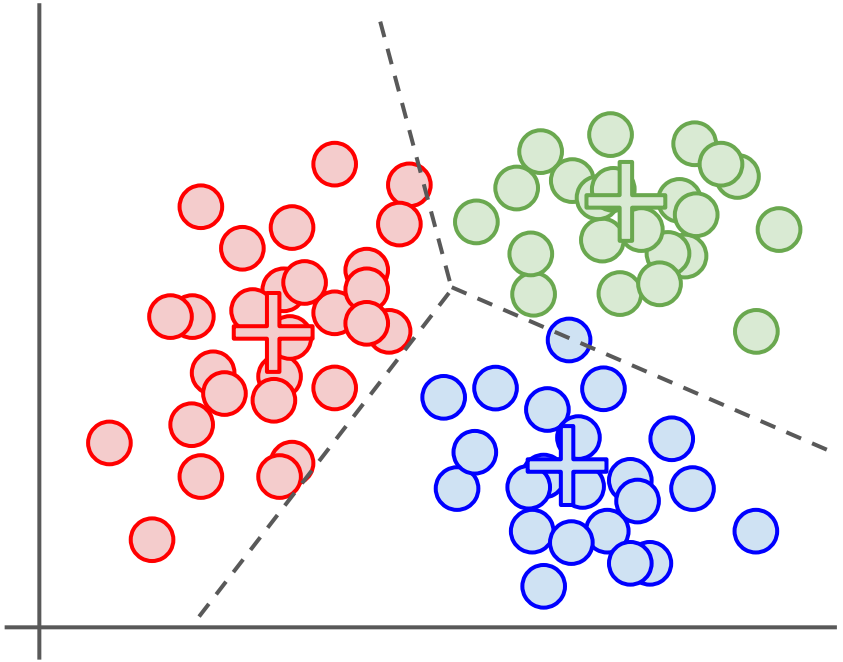
1. Define k (e.g. k=3)
2. Randomly choose centroid for each cluster
3. Assign every sample to nearest centroid based on Euclidean distance
4. Re-compute the centroid of the cluster
5. Repeat steps 3-4



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K-means Clustering

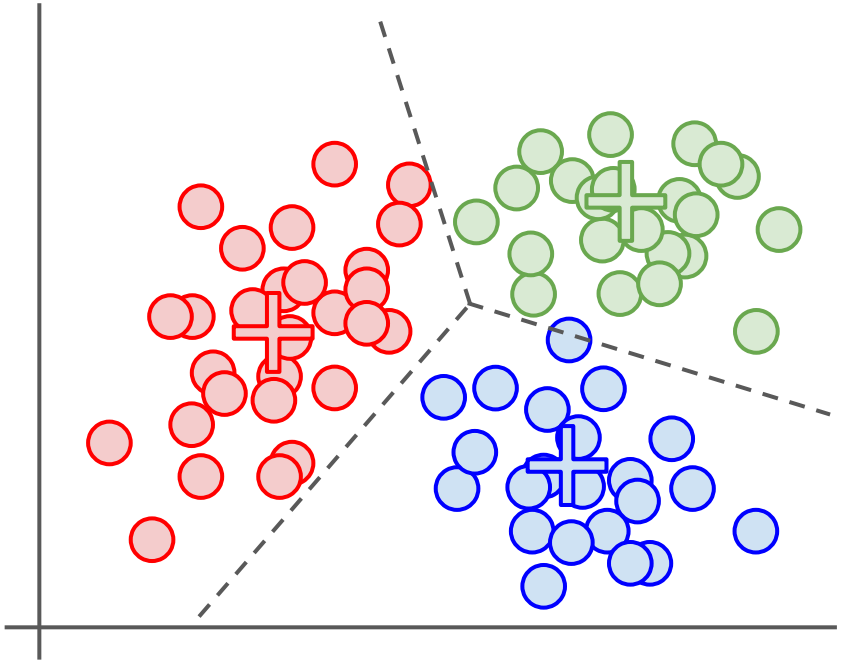
1. Define k (e.g. k=3)
2. Randomly choose centroid for each cluster
3. Assign every sample to nearest centroid based on Euclidean distance
4. Re-compute the centroid of the cluster
5. Repeat steps 3-4



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K-means Clustering

1. Define k (e.g. k=3)
2. Randomly choose centroid for each cluster
3. Assign every sample to nearest centroid based on Euclidean distance
4. Re-compute the centroid of the cluster
5. Repeat steps 3-4
6. …until one of:
    a. Sum of distances between data points and corresponding centroid is minimized
    b. No change in centroids
    c. Maximum iterations reached

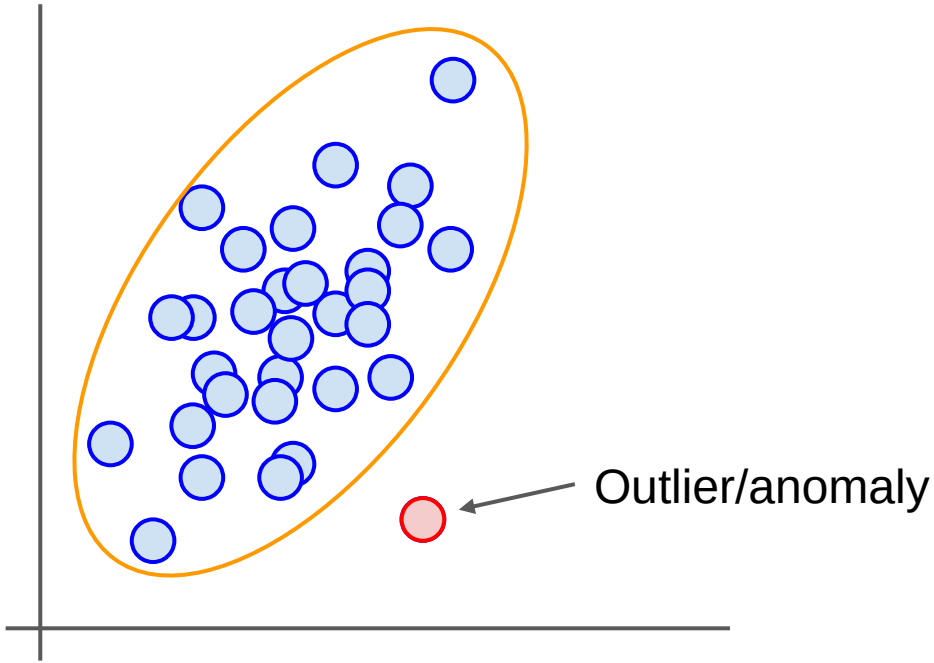


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Anomaly Detection

Examples:

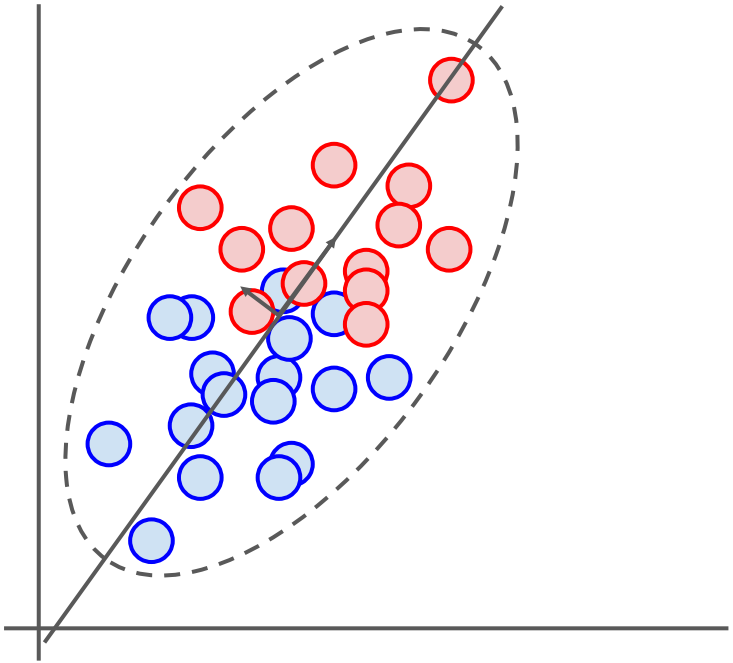
- Email spam
- Credit card fraud
- Motion alarm
- Fault detection

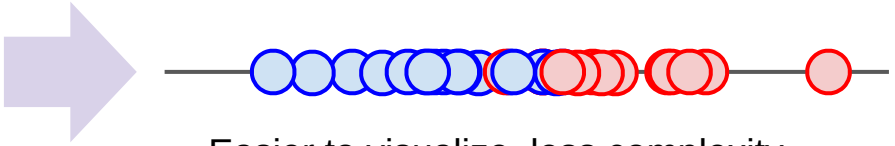


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Dimensionality Reduction

Example: principal component analysis (PCA)





Easier to visualize, less complexity

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